A Comprehensive Literature Survey on Voice-Based Detection of Parkinson's Disease Using Machine and Deep Learning

A Systematic Review

Review Period:	2020–2025
Final Search Date:	September 7, 2025
Studies Included:	69
Databases:	PubMed, Web of Science, IEEE Xplore, Scopus

Literature Survey Report

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1 Introduction and Review Context

1.1 Background

Parkinson's Disease (PD) is a prevalent and progressive neurodegenerative disorder and is generally regarded as the second most common neurodegenerative condition after Alzheimer's disease. The disease is associated with the gradual degeneration or dysfunction of dopamine-producing neurons in the substantia nigra. Dopamine is an important neurotransmitter involved in the regulation and coordination of movement.

The loss of dopaminergic neurons produces a range of motor and non-motor manifestations. The clinical presentation of PD is therefore heterogeneous and extends beyond movement-related symptoms.

1.1.1 Clinical Manifestations

Table 1: Major clinical manifestations associated with Parkinson's disease.

Symptom Category	**Specific Symptoms**
Motor Symptoms	Tremors, rigidity, bradykinesia, impaired balance and postural instability.
Non-Motor Symptoms	Cognitive impairment, depression, sleep disturbances and other neuropsychiatric symptoms.

1.1.2 Epidemiological and Diagnostic Context

Table 2: General epidemiological and diagnostic context of Parkinson's disease.

Statistic / Aspect	**Value / Description**
Global prevalence among individuals older than 60	Approximately 1%.
Dopaminergic neuron loss at motor onset	May reach substantial levels, with estimates of up to approximately 50%.
Traditional diagnosis	Clinical assessments, neurological examinations and observation of symptoms.
Diagnostic scales	Movement Disorder Society Unified Parkinson's Disease Rating Scale (MDS-UPDRS) and Hoehn and Yahr (H&Y) scale.

Traditional PD diagnosis is primarily clinical. Although clinical scales provide standardized ways of evaluating disease severity, diagnosis can be challenging, particularly during the early stages of the disease. This motivates the investigation of objective and easily accessible biomarkers.

1.2 The Promise of Artificial Intelligence in PD Detection

Recent developments in Artificial Intelligence (AI), cloud analytics and the Internet of Things (IoT) have created new opportunities for automated assessment of neurodegenerative diseases.

Table 3: Potential capabilities of AI-based PD monitoring systems.

Capability	**Description**
Real-Time Monitoring	Continuous or repeated assessment of symptoms outside conventional clinical environments.
Large-Scale Data Integration	Integration of heterogeneous information collected from multiple sources and devices.
Automated Clinical Decision Support	AI-assisted analysis that can support clinicians during screening and diagnostic decision-making.

1.2.1 Vocal Impairment as a Biomarker

Speech and voice abnormalities are among the important non-motor or communication-related manifestations associated with PD. A substantial proportion of individuals with PD experience some form of vocal impairment.

Table 4: Motivation for using voice as a biomarker for Parkinson's disease.

Aspect	**Detail**
Prevalence	Approximately 70–90% of individuals with PD may experience voice disorders.
Timing	Vocal changes may occur before prominent motor manifestations and have been reported as potentially preceding motor symptoms by several years.
Advantages	Non-invasive, relatively inexpensive, accessible and suitable for remote assessment.
Method	Voice can be recorded remotely using smartphones, computers, telephones and other digital recording devices.

The potential to obtain voice recordings remotely is particularly important for developing digital biomarkers. Unlike imaging or specialized neurological examinations, speech recordings can be collected using commonly available consumer devices.

1.2.2 Applications Beyond Parkinson's Disease

Speech-based machine learning and deep learning systems have also demonstrated potential for identifying other neurological and psychiatric conditions.

- **Depression:** Vocal tone, rhythm, prosody and speaking patterns can contain information related to depressive symptoms.
- **Emotional state:** Vocal characteristics can reflect changes in affective and psychological states.
- **Cognitive impairment:** Speech organization, pauses, lexical characteristics and acoustic patterns can provide information about cognitive decline.

These broader applications demonstrate the general potential of speech as a digital biomarker.

1.3 Evolution of Voice-Based PD Detection

1.3.1 Early Research: Classical Machine Learning

Early voice-based PD research primarily relied on manually designed acoustic features followed by conventional machine learning algorithms.

Table 5: Characteristics of classical voice-based PD detection approaches.

Approach	Techniques	Limitations
Handcrafted Features	MFCCs, jitter, shimmer and HNR	Requires manual feature design and may fail to capture complex representations.
Classical ML	SVM, Random Forest and KNN	Performance can be sensitive to recording conditions, dataset characteristics and feature quality.

1.3.2 Recent Advances: Deep Learning

Deep learning has changed the methodology by allowing representations to be learned automatically from acoustic signals.

Table 6: Major deep learning architectures used for voice-based PD detection.

Architecture	**Application**	**Advantages**
CNNs	Spectrogram and time-frequency analysis	Effective at learning local spectral patterns automatically.
RNNs/LSTMs	Temporal and sequential modeling	Can model temporal dependencies in speech.
Transformers	Speech representation learning and foundation models	Can capture long-range dependencies and provide strong transfer-learning capabilities.

The field is consequently moving from handcrafted feature engineering toward representation-learning frameworks.

1.4 Persistent Challenges

Despite considerable progress, several limitations remain.

Table 7: Persistent challenges in voice-based PD detection.

Challenge	**Description**
Dataset Heterogeneity	Differences in languages, recording protocols, demographic composition and speech tasks.
Small Sample Sizes	Many studies use fewer than 100 participants, increasing the risk of overfitting.
Lack of Standardization	Evaluation protocols, feature sets and speech tasks are not consistently defined.
Reproducibility	Incomplete methodological reporting and restricted datasets make replication difficult.
External Validation	Most studies rely primarily on internal validation, limiting evidence of real-world generalization.

1.5 Review Objectives

The major objectives of this systematic review are summarized below.

1. **Summarize** machine learning and deep learning approaches for voice-based PD detection between 2020 and 2025.
2. **Identify** commonly used datasets, speech tasks, feature extraction methods and model architectures.
3. **Evaluate** reported diagnostic performance, robustness and generalizability.
4. **Examine** how existing studies address dataset bias, validation and model transparency.

2 Review Methodology

2.1 Search Strategy

A systematic literature search was performed using four major scientific databases. The search covered publications from 2020 through 2025.

Table 8: Databases included in the literature search.

Database	**Search Period**
PubMed	2020–2025
Web of Science	2020–2025
IEEE Xplore	2020–2025
Scopus	2020–2025

The final search was conducted on September 7, 2025.

2.1.1 Search Terms

The search strategy combined terms related to Parkinson's disease, speech and voice, machine learning, artificial intelligence and detection.

Table 9: Search-term categories.

Category	Terms
PD	“Parkinson’s disease”, “Parkinsonism”, “PD”
Vocal	“speech”, “voice”, “vocal”, “voice signal”, “acoustic”
Methods	“machine learning”, “deep learning”, “artificial intelligence”, “AI”, “neural network”
Detection	“detect”, “diagnosis”, “classif”, “screen”

2.1.2 Filters Applied

The following filters were applied:

- Full-text articles.
- English-language publications.
- Journal articles.
- Publications within the defined 2020–2025 period.

2.2 Inclusion and Exclusion Criteria

2.2.1 Inclusion Criteria

Studies were included if they satisfied the following criteria:

- Investigated classification, diagnosis, detection or identification of PD.
- Applied machine learning or deep learning techniques.
- Used voice, speech or language-processing data.
- Were peer-reviewed, full-text and written in English.
- Bilingual or multilingual datasets were eligible when sufficient English documentation was available.

2.2.2 Exclusion Criteria

Table 10: Exclusion criteria used during study screening.

Category	Specific Exclusions
Non-Voice	Gait, EEG, handwriting and imaging biomarkers.
Non-ML/DL	Studies relying exclusively on conventional statistical methods.
Non-Original	Studies lacking sufficient methodological transparency or not presenting original methodological work.
Non-PD Focus	Studies concerning other neurological disorders without a relevant PD subgroup.
Incomplete	Studies missing essential information such as participant numbers, features or model configuration.
Non-Journal	Conference proceedings, reviews, meta-analyses, books and editorials.

2.3 Screening Process

The screening procedure followed the general structure of the PRISMA 2020 framework.

Table 11: Study-selection process.

Screening Stage	Records
Initial Identification	2,527
After Duplicate Removal	1,576
After Title/Abstract Screening	217
Full-Text Retrieval Attempted	217
Full Texts Inaccessible	99
Full Texts Available for Review	118
Final Included Studies	**69**

Full-text retrieval was attempted for 217 records. Ninety-nine records could not be accessed despite institutional and interlibrary search efforts. Consequently, 118 articles remained available for full-text evaluation, from which 69 studies met the final eligibility criteria.

2.3.1 Full-Text Exclusions

Table 12: Reasons for exclusion during full-text assessment.

Reason for Exclusion	**Count**
Lacked voice/speech data	9
Did not apply ML/DL	15
Insufficient methodological detail	9
Did not focus on PD	16
Total Excluded	**49**

2.4 Data Extraction

Data were extracted from the selected studies according to dataset characteristics, technical configuration, speech task, modeling strategy and evaluation procedure.

2.4.1 Dataset Information

Table 13: Dataset-related information extracted from the selected studies.

Aspect	**Details Collected**
Dataset Name	Source and collection procedures.
Demographics	Participant counts, age and gender.
Technical Information	Recording device, audio format and sampling rate.
Speech Tasks	Sustained vowels, reading, DDK repetition and other speech tasks.
Environment	Noise and recording conditions.
Diagnostic Tools	Clinical tools or criteria used to confirm PD.
Language	Language in which speech was recorded.
Availability	Public, restricted or private access.

2.4.2 Methodological Information

Table 14: Methodological information extracted from the selected studies.

Aspect	**Details Collected**
Author/Year	Publication information.
Task Type	Speech task employed by the study.
Dataset	Dataset name and characteristics.
Features	Acoustic, spectral, prosodic or learned features.
Method	Machine learning or deep learning model.
Evaluation	Cross-validation or train-test strategy.
Performance	Accuracy, sensitivity, specificity, F1-score and AUC.

2.5 Risk of Bias Assessment

Because the included literature primarily consisted of non-intervention machine learning studies, no formal intervention-oriented risk-of-bias instrument was applied. Instead, methodological quality was examined using indicators particularly relevant to machine learning research.

The assessment focused on:

- Dataset balance.
- Validation rigor.
- Potential subject-level data leakage.
- Appropriateness of reported evaluation metrics.
- Dataset size and representativeness.
- External validation.

Table 15: Major methodological concerns considered during the review.

Concern	How Addressed
Imbalanced datasets	Class imbalance and its effect on reported performance were discussed.
Small sample sizes	Dataset-size limitations and overfitting risks were examined.
Overfitting	Validation procedures were evaluated.
Metric overreliance	The limitations of accuracy-only reporting were considered.

3 Results

3.1 Annual Publication Trend

The number of publications increased progressively throughout the review period.

Table 16: Annual distribution of included studies.

Year	Number of Studies
2020	4
2021	6
2022	10
2023	12
2024	14
2025	23
Total	**69**

The publication trend shows a steady increase in research activity from 2020 to 2025, with the most notable growth occurring during 2023–2025. This increase reflects growing interest in speech-based digital biomarkers, machine learning and foundation-model approaches for PD detection.

3.2 Dataset Analysis

3.2.1 Most Frequently Used Datasets

Table 17: Frequently used datasets in voice-based PD research.

Dataset	Language	Studies	Description
PC-GITA	Spanish	19	Vowels, DDK, sentences, reading and monologues.
Istanbul PD (UCI)	Turkish	15	Sustained vowels; approximately 252 participants.
Oxford PD (UCI)	English	14	Sustained /a/ phonations; approximately 31 participants.
ItalianPVS	Italian	12	Vowels and phonetically balanced sentences.
Telephone PD (UAMS)	English	6	Sustained /a/ collected through telephone recordings.
MDVR-KCL	English	6	Spontaneous dialogue and read speech.
mPower	English	Variable	Smartphone /a/ phonations involving more than 5,800 participants.
PARK Framework	English	Variable	Pangram utterances involving approximately 1,306 participants.

3.2.2 Dataset Characteristics

Participant Size Distribution Most datasets contain fewer than 100 participants. Some datasets contain between 100 and 500 participants, while only a small number of resources provide more than 1,000 participants. Large-scale resources such as mPower and the PARK Framework represent important exceptions.

Table 18: General distribution of dataset sizes.

Size Range	Observation
Less than 100 participants	Most datasets
100–500 participants	Some datasets
More than 1,000 participants	Few datasets, including mPower and PARK Framework

Table 19: Approximate language distribution among identified datasets.

Language	**Number of Datasets**
English	7
Italian	5
Spanish	4
Chinese	3
Czech	3
French	1–2
German	1–2
Korean	1–2
Lithuanian	1–2
Portuguese	1–2
Turkish	1–2

Language Distribution The distribution demonstrates that English, Italian and Spanish are disproportionately represented relative to many other languages.

Gender Distribution Most datasets show a higher proportion of male participants. Female representation varies considerably between datasets.

Table 20: Observed demographic characteristics.

Observation	**Detail**
Male predominance	Many datasets contain a higher proportion of male participants, with male representation commonly around 50–60%.
Female representation	Female representation is more variable, often approximately 40–50%.
Possible explanation	The imbalance partly reflects the higher incidence of PD among men.
Age range	Participants are typically within approximately the 50–75 year age range.

3.2.3 Speech Task Distribution

Table 21: Distribution of major speech-task types.

Task Type	**Number of Datasets**
Vowel Phonation	21
Sentence Reading	16
DDK Repetition	10
Spontaneous Speech / Monologue	8
Picture Description	5
Conversation	4

Vowel phonation is the dominant speech task. Sustained vowels are particularly attractive because they are easy to record, require minimal linguistic content and can reveal several core acoustic abnormalities associated with PD.

3.2.4 Recording Condition Distribution

Table 22: Recording environments represented in the reviewed datasets.

Condition	**Number of Datasets**
Controlled clinical/laboratory	17
Real-world/mobile	8
Semi-controlled	4
Telephone-only	1

A gradual shift from controlled laboratory recordings toward real-world and mobile recordings can be observed. This transition is closely related to the increasing interest in remote health monitoring.

3.3 Feature Extraction Analysis

3.3.1 Input Type Distribution

Table 23: Major input representations used in the reviewed studies.

Input Type	Studies	Trend
Pre-computed acoustic features	33	Traditional and dominant approach.
Spectrograms / time-frequency representations	20	Increasing use with deep learning.
Speech embeddings / foundation models	10	Rapid adoption, particularly from 2023 onward.
Raw audio	10	End-to-end deep learning approaches.
Text / linguistic features	2	Emerging direction.
Synthetic / augmented data	2	Generative and data-augmentation approaches.

3.3.2 Handcrafted Feature Categories

Table 24: Common handcrafted acoustic feature categories.

Feature Category	Examples	Extraction Toolkits
Perturbation	Jitter, shimmer, HNR and NHR	Praat, openSMILE, DisVoice
Nonlinear	RPDE, DFA and PPE	Praat and custom implementations
Spectral	MFCCs, formants and RASTA-PLP	openSMILE and Praat
Prosodic	Pitch, intensity and duration	Praat and openSMILE

3.3.3 Common Feature Processing Techniques

Table 25: Feature-processing methods frequently used in PD detection studies.

Technique	**Purpose**
Recursive Feature Elimination (RFE)	Selection of informative features while reducing dimensionality.
Principal Component Analysis (PCA)	Dimensionality reduction and transformation into a lower-dimensional feature space.
SMOTE	Synthetic oversampling of minority classes.
Feature Scaling	Normalization of feature magnitudes before machine learning.

3.3.4 Foundation Models

The increasing availability of large-scale pretrained speech models has introduced a new paradigm for voice-based PD detection.

Table 26: Representative speech foundation models used in PD-related research.

Model	**Type**	**Key Feature**
Wav2Vec 2.0	Self-supervised	Learns speech representations through self-supervised pretraining.
HuBERT	Self-supervised	Uses masked prediction to learn contextual speech representations.
Whisper	Weakly supervised	Large-scale speech recognition pretraining.
WavLM	Self-supervised	Enhanced contextual speech representation learning.

The adoption of foundation models has increased substantially during 2023–2025. These models can reduce dependence on manually engineered acoustic features and may provide representations that transfer across datasets and languages.

3.4 Model Architecture Analysis

3.4.1 Overall Distribution

Table 27: Distribution of major modeling approaches.

Model Type	Studies	Percentage
Classical ML Only	28	40.6%
Deep Learning Only	29	42.0%
Hybrid ML + DL	12	17.4%

The distribution indicates that the field is undergoing a transition. Classical machine learning and deep learning approaches occur at broadly comparable frequencies, while hybrid methods represent a smaller but important group.

3.4.2 Classical Machine Learning Models

Table 28: Common classical machine learning models.

Model	Prevalence	Common Features / Characteristics
SVM	Most common	Often combined with RBF kernels and feature selection.
Random Forest	Highly used	Ensemble of decision trees based on impurity or related splitting criteria.
XGBoost / LightGBM	Growing	Gradient-boosting frameworks suitable for structured acoustic features.
KNN	Moderate	Distance-based classification.
Logistic Regression	Moderate	Regularized linear classification.

Classical models frequently report accuracy values above 90% on controlled datasets. However, these results can deteriorate substantially when recording conditions or populations change.

3.4.3 Deep Learning Models

Table 29: Major deep learning architectures used in the reviewed literature.

Architecture	**Studies**	**Applications**
CNNs	12	Spectrogram analysis and automatic pattern recognition.
Transformer / Foundation Models	11	Self-supervised representation learning and cross-lingual modeling.
RNN / Hybrid	Fewer	Temporal and sequential speech modeling.
Autoencoder / GAN	Fewer	Representation learning and data augmentation.

3.4.4 Detailed Deep Learning Distribution

Table 30: Detailed distribution of deep learning and hybrid architectures.

Model Type	**Number**
CNN	12
Transformer / Foundation	11
RNN / LSTM	8
Autoencoder / GAN	3
Hybrid	12

Several studies employ more than one architecture. Therefore, the architecture counts should not necessarily be interpreted as mutually exclusive categories.

3.4.5 Hybrid Approaches

Hybrid approaches combine learned representations with conventional classifiers or ensemble strategies.

Table 31: Examples of hybrid modeling strategies.

Approach	Example	Key Finding
DL feature extraction + ML classification	CNN + Random Forest	Combines automatic representation learning with a conventional classifier.
Ensemble methods	Multiple models	Can improve robustness by combining complementary predictions.
Cascade models	SVM followed by DNN	Sequential feature refinement and classification.

Some hybrid studies report accuracy values approaching 99–100% on small datasets. Such results should be interpreted cautiously because small samples and non-independent validation can lead to optimistic estimates.

3.5 Validation Strategy Analysis

3.5.1 Validation Strategy Distribution

Table 32: Validation strategies reported across the literature.

Strategy	Studies	Percentage
k-Fold Cross-Validation	28	∼40%
Fixed Train/Test Split	25	∼36%
External / Federated	9	∼13%
Subject-Independent (LOSO/LOO)	8	∼12%
Nested / Multi-Stage	3	∼4%
Repeated / Aggregated	1	∼1%

Internal validation dominates the reviewed literature. Approximately 76% of studies rely primarily on internal validation approaches, whereas external validation is uncommon.

Subject-independent evaluation is particularly important for speech-based PD detection because multiple recordings from the same speaker can otherwise appear in both training and testing partitions. Such overlap can produce artificially high performance.

3.5.2 Performance Metrics

Table 33: Common evaluation metrics.

Metric	Prevalence	Utility
Accuracy	Most common	Measures overall proportion of correctly classified samples.
F1-score	Common	Balances precision and recall.
AUC	38 studies	Measures discrimination across classification thresholds.
Sensitivity	Common	Measures the ability to detect PD cases.
Specificity	Common	Measures the ability to correctly identify healthy controls.

3.5.3 Reported Performance Ranges

Table 34: Reported performance ranges across the reviewed studies.

Metric	Reported Range
Accuracy	68.56%–100%
AUC	0.73–1.00

Most reported AUC values fall approximately between 0.85 and 0.95. Nevertheless, high internal performance should not automatically be interpreted as evidence of clinical readiness.

3.6 Performance by Model Type

The reviewed literature indicates that classical machine learning, deep learning and hybrid models can all achieve high performance on controlled datasets.

Table 35: General performance characteristics by model type.

Model Type	Median Accuracy	Observed Spread
Classical ML	$> 90\%$	High variability.
Deep Learning	$> 90\%$	High variability.
Hybrid	$> 90\%$	High variability.

The apparent similarity between model families suggests that the validation protocol and dataset characteristics may be at least as important as the choice of classifier.

A major concern is publication bias: studies reporting high performance may be more likely to be published. Consequently, the literature-level median accuracy may overestimate the performance expected in independent clinical populations.

3.7 Interpretability Analysis

Interpretability remains an important but underdeveloped component of voice-based PD detection.

Table 36: Interpretability approaches reported in the literature.

Method	**Use**	**Purpose**
SHAP	Few studies	Estimates feature contributions to predictions.
Grad-CAM	Few studies	Provides spatial localization for image-like representations.
Cross-Attention	Few studies	Provides information about interactions between learned representations.

Although interpretability techniques can provide useful explanations, further work is needed to determine whether model explanations correspond to clinically meaningful vocal characteristics.

4 Critical Analysis and Discussion

4.1 Dataset Design and Accessibility

4.1.1 Trade-Offs in Dataset Design

Different datasets optimize different objectives.

Table 37: Trade-offs involved in voice-dataset design.

Priority	**Design Choice**	**Trade-Off**
Clinical Depth	Rich protocols such as PC-GITA, FraLusoPark and NeuroVoz	Typically associated with smaller sample sizes.
Scalability	Simple tasks such as those used in mPower and PARK Framework	May provide less detailed vocal information.
Real-World Relevance	Mobile and telephone recordings	Introduces greater acoustic variability.
Scientific Control	Clinical or laboratory recordings	May reduce real-world generalizability.

This trade-off demonstrates that there is no single ideal dataset design. A dataset with highly controlled recordings may provide cleaner signals but may not adequately represent the conditions encountered during real-world use.

4.1.2 Dataset Accessibility

Table 38: Dataset accessibility characteristics.

Accessibility Status	**Examples**	**Issue**
Publicly Accessible	UCI datasets, NeuroVoz, ItalianPVS	Useful for reproducibility but often limited by legacy collection protocols.
Restricted / Private	PC-GITA, FraLusoPark and some Chinese datasets	Limits independent replication and cross-study comparisons.
Feature-Only	Some datasets, including certain Lithuanian resources	Prevents researchers from re-engineering or validating the feature extraction pipeline.

Restricted data accessibility represents an important bottleneck for reproducibility. If researchers can access only precomputed features rather than raw audio, differences in preprocessing and feature extraction cannot be independently evaluated.

4.1.3 Demographic and Linguistic Bias

Table 39: Potential sources of demographic and linguistic bias.

Bias	**Potential Impact**
Gender	Male-skewed datasets may reduce performance when models are applied to female speakers.
Age	Narrow age distributions may limit generalization to different age groups.
Language	Dominance of English, Spanish and Italian limits evidence for cross-lingual generalization.
Geography	Concentration in North American and European populations limits global representativeness.

4.2 Advancements in Machine Learning

4.2.1 The ML-to-DL Transition

The transition from classical machine learning to deep learning can be summarized as follows.

Table 40: Comparison between classical ML, deep learning and hybrid approaches.

Aspect	**Classical ML**	**Deep Learning**	**Hybrid**
Features	Handcrafted acoustic features	Automatically learned representations	DL-derived representations plus conventional features/classifiers
Models	SVM, RF, Logistic Regression	CNNs, RNNs, Transformers	CNN + RF, SVM + DNN and ensembles
Strengths	Efficient and effective with small structured datasets	Can capture complex patterns	Combines representation learning with conventional classification
Weaknesses	Dependent on feature engineering and recording conditions	Data-hungry and potentially less interpretable	Greater model complexity and computational requirements

Classical machine learning remains useful when datasets are small and feature engineering is well understood. Deep learning, however, provides the advantage of learning representations directly from signals or time-frequency representations.

Hybrid approaches attempt to exploit the strengths of both paradigms.

4.3 Foundation Models: The New Frontier

Large pretrained speech models represent one of the most significant developments in recent voice-based PD research.

4.3.1 Advantages of Foundation Models

Table 41: Potential advantages of foundation speech models.

Advantage	**Description**
Self-Supervised Pretraining	Models can learn from large quantities of speech without requiring manual labels for every training example.
Reduced Feature Engineering	Learned representations reduce dependence on manually selected acoustic features.
Cross-Lingual Potential	Pretraining on diverse speech data may support transfer across languages.
Few-Shot / Zero-Shot Potential	Large pretrained representations may reduce the amount of labeled clinical data required for adaptation.

4.3.2 Representative Performance Examples

Table 42: Examples of recent foundation-model studies.

Study	**Model**	**Dataset**	**Performance**
Dao et al. (2025)	Ensemble of Wav2Vec2, Whisper and SeamlessM4T	ICEBERG	AUC 91.35%.
Sedigh Malekroodi et al. (2025)	Wav2Vec2 + contrastive learning	NeuroVoz	AUC 0.92.

4.3.3 Limitations for Clinical Use

Table 43: Clinical limitations of foundation speech models.

Limitation	Description
Interpretability	Learned embeddings do not necessarily correspond directly to clinically meaningful vocal characteristics.
Computational Cost	Large pretrained models may require substantial computational resources.
Domain Adaptation	Pathological speech differs from the speech distributions used during pretraining.
Fine-Tuning Challenges	Small clinical datasets make stable and reliable fine-tuning difficult.

4.4 Methodological and Reporting Limitations

4.4.1 Validation Strategy Issues

Table 44: Major validation-related problems.

Issue	Impact
Internal validation dominance	May produce optimistic performance estimates.
Limited external validation	Leaves uncertainty regarding performance in independent populations.
Subject-dependent splits	Can result in speaker leakage and inflated metrics.
Inconsistent reporting	Makes comparison between studies difficult.

4.4.2 Cross-Dataset Generalization

One of the most important observations from the literature is the gap between within-dataset and cross-dataset performance.

Table 45: Illustrative cross-dataset performance difference.

Evaluation Condition	**Performance**
Within Dataset	$> 90\%$ accuracy in many studies
Cross Dataset	Approximately 33–74% accuracy in reported examples

This large decrease demonstrates that high within-dataset performance does not necessarily imply robust clinical generalization.

4.4.3 Demographic Bias

Only a small number of studies explicitly address demographic bias.

Table 46: Examples of demographic-bias investigations.

Bias	**Studies / Observations**
Gender	Addressed by only a limited number of studies, including Momeni et al. (2025).
Age	Explicit analysis remains uncommon.
Language	Some studies investigate multilingual generalization, including Favaro et al. (2023).

Momeni et al. (2025) reported an improvement of approximately 9.5 percentage points using group-wise normalization accounting for age and gender, illustrating that demographic normalization can have a meaningful influence on model performance.

4.4.4 Class Imbalance

Class imbalance is another important methodological concern.

Table 47: Examples of class-imbalance problems.

Issue	Example	Potential Impact
Skewed PD:HC ratio	Istanbul dataset: approximately 188 PD versus 64 HC	Accuracy may hide poor minority-class performance.
SMOTE oversampling	Frequently used in imbalanced datasets	May alter the training distribution and potentially produce optimistic estimates if improperly applied.
Low specificity	Example: Hawi et al. (2022), approximately 71.1% specificity	Can result in a high number of false positives.

Accuracy alone is therefore insufficient when class distributions are unequal. Sensitivity, specificity, precision, F1-score and AUC provide complementary information.

4.4.5 Dataset Usage Patterns

Table 48: Patterns of dataset usage.

Dataset Use	Studies	Interpretation
Single Dataset	∼65%	Limited evidence of generalizability.
Two Datasets	10	Provides stronger evidence than single-dataset evaluation.
Three Datasets	7	Enables broader comparison.
Four or More Datasets	6	Represents more rigorous cross-dataset validation.

A recurring observation is that models reporting greater than 90% accuracy on a single dataset may exhibit substantial degradation when evaluated on an independent dataset.

4.5 Gaps Identified

4.5.1 Data Gaps

The following data-related gaps are evident:

1. **Limited multilingual representation:** Existing datasets are disproportionately concentrated in English, Spanish and Italian.
2. **Demographic bias:** Many datasets are male-skewed and contain limited age diversity.
3. **Restricted access:** Several valuable datasets remain private or restricted.
4. **Absence of standardized benchmarks:** There is no universally adopted benchmark for voice-based PD detection.

4.5.2 Methodological Gaps

1. Evaluation protocols are inconsistent.
2. Subject-independent validation is not universally adopted.
3. External validation remains uncommon.
4. There is no consensus regarding the optimal feature set.
5. Speech tasks vary substantially between studies.

4.5.3 Reporting Gaps

1. Confidence intervals are frequently absent.
2. Variance and standard deviations are rarely reported.
3. Subgroup analyses by gender and age are uncommon.
4. Interpretability analyses are limited.

4.6 Comparison with Previous Systematic Reviews

Table 49: Comparison with related systematic reviews.

Review	Focus	Key Finding
Current Review (2025)	Voice-based ML/DL, 2020–2025	Field is transitioning toward deep learning and foundation models.
Ngo et al. (2022)	Speech and voice for PD	Systematic review of computerized speech and voice analysis.
Rabie et al. (2025)	ML/DL for PD	Covers multiple PD modalities.
Altham et al. (2024)	Cognitive impairment in PD	Speech represents one potential assessment modality.

4.6.1 Unique Contribution of the Current Review

Table 50: Distinctive characteristics of the current review.

Aspect	Contribution
Scope	Voice-specific review covering 2020–2025.
Comprehensiveness	Analysis of datasets, features, models and validation strategies.
Quantitative Analysis	Frequency distributions and performance ranges.
Critical Analysis	Assessment of demographic bias, methodological limitations and generalization gaps.
Future Directions	Actionable recommendations for dataset, modeling and clinical research.

5 Future Directions

5.1 Standardization Needs

Standardization is essential for meaningful comparison between future systems.

Table 51: Recommended areas for methodological standardization.

Area	**Recommendation**
Evaluation Protocols	Adopt subject-independent and external validation whenever possible.
Metrics	Report AUC, F1-score, sensitivity and specificity rather than accuracy alone.
Task Definition	Develop common speech tasks for improved comparability.
Feature Sets	Define reproducible feature-extraction pipelines.
Reporting	Provide confidence intervals, variance and subgroup analyses.

5.2 Dataset Development

Table 52: Recommended priorities for future dataset development.

Priority	**Recommended Action**
Large Multilingual Datasets	Collect speech from diverse linguistic and cultural populations.
Open Access	Provide datasets whenever ethical, legal and privacy constraints allow.
Demographic Diversity	Balance representation across gender, age and disease stage.
Real-World Conditions	Include mobile, telephone and naturally occurring recording conditions.
Longitudinal Data	Track disease progression over time.

5.3 Model Development

Table 53: Recommended directions for future model development.

Priority	Recommended Action
Foundation Models	Perform domain adaptation for pathological speech.
Interpretability	Use SHAP, Grad-CAM and attention visualization where appropriate.
Hybrid Approaches	Combine learned representations with conventional classifiers where beneficial.
Fairness-Aware Modeling	Explicitly address demographic differences and subgroup performance.
Multi-Modal Integration	Combine speech with gait, handwriting and other digital biomarkers.

5.4 Clinical Translation

For voice-based PD detection to become clinically useful, high experimental accuracy alone is insufficient.

Table 54: Priorities for clinical translation.

Priority	Recommended Action
Prospective Validation	Evaluate systems prospectively in clinical populations.
Regulatory Pathways	Consider relevant FDA, EMA and other regulatory requirements.
Interpretability	Provide explanations that clinicians can understand and evaluate.
Integration	Integrate AI systems with electronic health records and clinical workflows.
At-Home Monitoring	Develop secure mobile and remote-monitoring applications.

6 Limitations of the Current Review

The review itself has several limitations that should be considered when interpreting its conclusions.

Table 55: Limitations of the current systematic review.

Limitation	Description
No Pre-Registered Protocol	The review did not use a pre-registered protocol, which may affect reproducibility.
Binary Classification Focus	The review focuses primarily on PD versus healthy-control detection and gives limited attention to staging and progression tracking.
English-Only Inclusion	English-language restrictions may exclude relevant non-English literature.
No Formal Meta-Analysis	High heterogeneity between datasets, tasks and validation strategies prevents meaningful quantitative synthesis.
Database Limitations	Only four databases were searched.
Publication Bias	High-performing studies may have a greater probability of publication.
Sustained Phonation Emphasis	The strong representation of sustained vowels may not capture the complexity of spontaneous speech.

7 Conclusion

7.1 Summary of Key Findings

The literature demonstrates substantial progress in voice-based Parkinson's disease detection, but it also reveals significant limitations that prevent direct translation of reported research performance into clinical reliability.

1. **Field in Transition:**

   Classical machine learning represents approximately 40.6% of the reviewed studies, deep learning approximately 42.0%, and hybrid approaches approximately 17.4%. This distribution reflects the ongoing transition from handcrafted acoustic features toward learned representations and foundation models.

2. **High Performance on Controlled Data:**

   Many studies report accuracy above 90% on relatively small and homogeneous datasets. Classical ML and deep learning models can both achieve high performance under controlled experimental conditions.

3. **Generalization Gap:**

Performance often decreases substantially when models are evaluated on independent datasets. Reported cross-dataset accuracy can fall into the range of approximately 33–74%, demonstrating that within-dataset performance does not guarantee real-world generalization.

4. **Dataset Issues:**

   Many datasets contain fewer than 100 participants and exhibit demographic and linguistic imbalances. English, Spanish and Italian are disproportionately represented, while several valuable datasets remain restricted.

5. **Validation Issues:**

   Internal validation dominates the literature, accounting for approximately 76% of studies. External validation remains relatively rare, occurring in fewer than 15% of studies.

6. **Interpretability Gap:**

   Most high-performing deep learning and foundation-model systems remain difficult to interpret. Clinical adoption requires evidence that the learned representations correspond to meaningful and reproducible vocal biomarkers.

7.2 Contributions to the Field

Table 56: Major contributions synthesized by the review.

Contribution	**Significance**
Comprehensive Synthesis	Analysis of 69 studies published during 2020–2025.
Dataset Analysis	Characterization of approximately 29 datasets.
Model Analysis	Comparison of classical ML, deep learning and hybrid approaches.
Validation Analysis	Assessment of internal, subject-independent and external validation strategies.
Gap Identification	Identification of demographic bias, validation limitations, accessibility problems and interpretability gaps.
Future Directions	Development of actionable recommendations for future research.

7.3 Final Remarks

Voice-based Parkinson's disease detection has reached a high level of technical maturity in terms of the number and sophistication of available machine learning and deep learning

methods. However, technical performance should not be confused with clinical readiness.

The central challenge is increasingly shifting from *"Can a model classify PD accurately within a dataset?"* to *"Can the model reliably detect PD in an unseen population under realistic recording conditions?"*

Future research should therefore prioritize real-world robustness, demographic fairness, reproducibility and clinical utility rather than simply maximizing accuracy on a single dataset.

Large multilingual datasets, standardized evaluation protocols, subject-independent validation, external validation, interpretable modeling and prospective clinical studies are required to bridge the gap between research prototypes and deployable clinical systems.

7.3.1 Critical Need

The field would benefit substantially from community-wide adoption of standardized evaluation protocols that include:

- Subject-independent and external validation.
- Transparent reporting of sensitivity-specificity trade-offs.
- Subgroup analyses by gender, age and language.
- Confidence intervals and measures of variance.
- Open-access benchmarks and shared evaluation tasks.
- Explicit reporting of preprocessing and feature-extraction pipelines.
- Independent cross-dataset evaluation.

Ultimately, voice-based PD detection has the potential to become a scalable digital biomarker for screening and remote monitoring. Realizing this potential will require a coordinated shift from isolated high-accuracy experiments toward rigorous, reproducible and clinically meaningful evaluation.

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